

INTERVIEW PREPARATION · ROUTING & SWITCHING

The CCNA Interview Handbook

Two hundred and ten questions a hiring manager can actually ask you — each with an answer you can say out loud in under a minute, and a note on what the question is really testing.

20

TOPIC BLOCKS

210

QUESTIONS

CLI

VERIFIED COMMANDS

L1—L4

COVERAGE

How to use this handbook

Most candidates fail networking interviews for one of two reasons: they memorized a definition without understanding the mechanism behind it, or they knew the mechanism but couldn't explain it in plain language while someone watched them. This handbook is built to fix both problems.

Every entry gives you a short, accurate answer — the kind you can deliver in thirty to sixty seconds. Where a question is commonly used as a filter, you'll also find a **What they're checking** note that tells you what the interviewer is listening for and where candidates usually slip.

SAY IT, DON'T READ IT

Work through one topic block per session and answer out loud before you look. Reading feels like progress; speaking is the actual skill being tested.

LEAD WITH THE ONE-LINER

Open with a single sentence of definition, then add mechanism, then an example. Interviewers stop you when they've heard enough — so put the answer first.

PROVE IT IN THE LAB

Every command in this book runs on Cisco IOS. Type them in Packet Tracer, CML, or EVE-NG. A candidate who has seen the output answers differently from one who hasn't.

SAY "I DON'T KNOW" WELL

Guessing is punished harder than a gap. Name what you do know, state the boundary, and say how you'd find out. That reads as senior, not weak.

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Networking Fundamentals

The warm-up round. These questions look easy, which is exactly why they are used to separate candidates who understand networks from candidates who have memorized a glossary. Precision matters more than length here.

1.1 What is a computer network, and what problem does it solve?

A network is a set of devices connected by a shared transmission medium that agree on a common set of rules — protocols — so they can exchange data. The problem it solves is resource sharing: files, applications, printers, internet access, and compute capacity become reachable from anywhere instead of being locked to one machine.

A useful framing for an interview: a network has three ingredients — endpoints that produce and consume data, media that carry it, and protocols that make the exchange predictable. Remove any one of the three and you no longer have a network.

1.2 What is the difference between a collision domain and a broadcast domain?

A **collision domain** is a segment where two devices transmitting at the same instant will corrupt each other's frames. A **broadcast domain** is the set of devices that will receive a frame sent to the broadcast address ff:ff:ff:ff:ff:ff.

The device boundaries are what interviewers want: a hub extends both domains; every *switch port* is its own collision domain, and every *VLAN* is its own broadcast domain; a router terminates broadcast domains because it does not forward Layer 2 broadcasts.

WHAT THEY'RE CHECKING

On a modern full-duplex switched network, collisions should be zero. If you can add that collision domains are essentially a legacy concern — and that seeing collisions today usually means a duplex mismatch or a hub someone forgot about — you have separated yourself from the textbook answer.

1.3 Explain the difference between a hub, a switch, and a router.

DEVICE	LAYER	FORWARDING DECISION	DOMAINS
Hub	1	None — repeats the electrical signal out every other port	One collision domain, one broadcast domain
Switch	2	Destination MAC address, looked up in the MAC address table	Collision domain per port, broadcast domain per VLAN
Router	3	Destination IP address, longest-prefix match in the routing table	Broadcast domain per interface

A switch also floods a frame out all ports in the VLAN when the destination MAC is unknown, when it is a broadcast, or when it is an unknown multicast — the three cases collectively called BUM traffic. That flooding behavior is why a switch is not simply a "fast hub."

1.4 Define bandwidth, throughput, latency, and jitter.

- **Bandwidth** — the theoretical maximum data rate of a link, in bits per second. A property of the link.
- **Throughput** — the rate actually achieved, after protocol overhead, congestion, retransmissions, and the slowest hop on the path.
- **Latency** — the one-way delay for a bit to travel from source to destination; round-trip time (RTT) is what ping reports.
- **Jitter** — the variation in latency between successive packets.

Real-time traffic cares about the last two. A voice call is unusable at 400 ms of jitter even on a 10 Gbps link, while a file transfer barely notices.

WHAT THEY'RE CHECKING

Whether you conflate bandwidth with speed. Adding bandwidth does not reduce latency — propagation delay is bounded by distance and the speed of signal in the medium.

1.5 How does a MAC address differ from an IP address, and why do we need both?

A MAC address is a 48-bit Layer 2 identifier burned into the NIC — the first 24 bits are the vendor OUI. It is flat, has no structure that can be aggregated, and is only meaningful inside a single broadcast domain. An IP address is a 32-bit (IPv4) or 128-bit (IPv6) Layer 3 identifier that is hierarchical, assigned by the network, and therefore routable across the world.

We need both because forwarding happens at two scales. The IP header answers "where is this ultimately going," and stays unchanged end to end. The Ethernet header answers "who is the next device on this wire," and is rewritten at every router hop.

WHAT THEY'RE CHECKING

Say explicitly that the source and destination IP stay the same across the path while the source and destination MAC change at every hop. That single sentence is the most common make-or-break moment in an entry-level interview.

1.6 Define unicast, broadcast, multicast, and anycast.

- **Unicast** — one sender, one specific receiver.
- **Broadcast** — one sender, every host in the broadcast domain. Exists in IPv4 (255.255.255.255 and the subnet broadcast), and does not exist in IPv6.
- **Multicast** — one sender, a group of interested receivers that have subscribed to a group address (224.0.0.0/4 in IPv4).
- **Anycast** — one address advertised from many locations; routing delivers the traffic to the topologically nearest instance. This is how public DNS resolvers and CDNs work.

1.7 What is the difference between a public and a private IP address?

Public addresses are globally unique, allocated through IANA and the regional registries, and routable on the internet. Private addresses are defined by RFC 1918 — 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16 — and are dropped by internet routers, so the same range can be reused inside thousands of organizations. NAT translates between the two.

Worth knowing alongside them: 169.254.0.0/16 is link-local (APIPA), which a host self-assigns when DHCP fails; 127.0.0.0/8 is loopback; and 100.64.0.0/10 is carrier-grade NAT space.

1.8 Why do we subnet?

Three reasons, in order of how often they actually drive the decision:

- **Containment** — a broadcast domain has a practical size limit. Splitting a flat /16 into /24s keeps broadcast and flooded traffic local.
- **Security and policy** — subnets give you a boundary to apply access lists, firewall rules, and QoS to.
- **Address efficiency** — VLSM lets you give a point-to-point WAN link a /30 instead of wasting a /24 on two hosts.

1.9 What is a default gateway, and what happens if a host has the wrong one?

The default gateway is the router interface a host sends traffic to when the destination is not on its own subnet. The host compares the destination IP against its own address and mask; if the network portions differ, it ARPs for the gateway's MAC and sends the frame there with the original destination IP intact.

With a wrong gateway, local communication inside the subnet still works perfectly — which is the classic symptom pattern. The user can reach their neighbor and the local file server but nothing off-subnet, and ping to the gateway itself fails.

1.10 When do you use a straight-through cable versus a crossover cable?

Traditionally, straight-through connects unlike devices (PC to switch, switch to router) and crossover connects like devices (switch to switch, PC to PC, router to router). In a straight-through cable both ends use the same pinout; in a crossover, pins 1 and 3 and pins 2 and 6 are swapped so that transmit lands on receive.

The honest modern answer is that **auto-MDIX** on any current switch detects and corrects the crossover electronically, so cable type rarely matters. It still matters when auto-MDIX is disabled, when the port is hard-coded to a speed and duplex (auto-MDIX requires autonegotiation), or on older gear.

1.11 What is a duplex mismatch and how would you detect one?

It happens when one side is set to full duplex and the other to half — most often because one end is hard-coded and the other is left on autonegotiation, which then falls back to half. The link comes up and pings succeed, but throughput collapses under load.

You detect it with `show interfaces`: the half-duplex side reports late collisions, and the full-duplex side reports FCS errors and runs. The fix is to make both ends match — ideally autonegotiation on both, or hard-coded on both.

WHAT THEY'RE CHECKING

This is a favorite because it is a real-world fault that a book-only candidate has never seen. The phrase "the link is up but slow" should immediately make you mention duplex.

1.12 CSMA/CD versus CSMA/CA — what is the difference and why are there two?

Both are ways of sharing one medium. **CSMA/CD** (Collision Detection) is the legacy Ethernet method: transmit, and if you detect a collision, send a jam signal and back off for a random interval before retrying. **CSMA/CA** (Collision Avoidance) is the 802.11 wireless method: wait for a clear channel, wait an additional random backoff, then transmit, and rely on an explicit acknowledgment to confirm delivery.

Wireless needs avoidance rather than detection because a station cannot reliably hear a collision while it is transmitting, and because of the hidden node problem — two stations may both hear the access point but not each other. Optional RTS/CTS handshaking addresses that case.

1.13 A packet arrives at a router. Walk me through how it chooses a route.

Three tests, always in this order:

- **Longest prefix match first.** Among all routes that contain the destination, the most specific mask wins. A /32 host route beats a /24, which beats the 0.0.0.0/0 default. This decision ignores protocol and metric entirely.
- **Administrative distance** breaks ties between different sources offering the *same* prefix — a static /24 (AD 1) beats an OSPF /24 (AD 110).
- **Metric** breaks ties within the same protocol. Equal-cost paths are installed together and load-balanced.

If nothing matches and there is no default route, the router drops the packet and returns an ICMP destination-unreachable message.

WHAT THEY'RE CHECKING

Candidates almost always say "administrative distance" first. Getting the order right — specificity, then AD, then metric — is the single strongest signal you actually understand route selection.

1.14 What is administrative distance? Give me some values.

Administrative distance is a locally significant number expressing how much a router trusts the *source* of a route. It is never advertised to neighbors, and lower is better.

SOURCE	AD	SOURCE	AD
Connected	0	OSPF	110
Static	1	RIP	120
eBGP	20	External EIGRP	170
Internal EIGRP	90	iBGP	200

An AD of 255 means the route is considered unusable and will never be installed. Raising the AD of a static route deliberately — a *floating static* — is the standard way to build a backup path that only appears when the dynamic route disappears.

1.15 Static or dynamic routing — how do you decide?

Static routing is deterministic, adds no CPU or bandwidth overhead, and is not discoverable by an attacker probing for routing protocols. It does not scale and does not react to a topology change on its own.

Dynamic routing scales, converges automatically around failures, and is what you want anywhere the topology can change. It costs CPU, memory, and bandwidth, and it introduces a trust relationship that has to be secured with neighbor authentication.

In practice: statics for stub sites, a single-homed branch, and the default route toward the ISP; a dynamic protocol everywhere the topology has more than one path.

1.16 Name the memory types in a Cisco router and what each one holds.

- **ROM** — bootstrap code, POST diagnostics, and the ROMMON recovery environment. Not user-writable.
- **Flash** — the IOS image itself, and usually a backup config. Non-volatile.
- **RAM** — the running configuration, routing table, ARP cache, and packet buffers. Volatile: lost on reload.
- **NVRAM** — the startup configuration, which IOS reads at boot. Non-volatile.

This is why `copy running-config startup-config` exists — it moves your work from RAM into NVRAM. The boot order itself is POST, then bootstrap from ROM, then locate and load IOS from flash, then load the startup config from NVRAM.

1.17 Telnet or SSH?

SSH, without qualification. Telnet (TCP 23) sends everything, credentials included, in cleartext, so a single packet capture on any intermediate segment yields the enable password. SSH (TCP 22) authenticates the device and encrypts the session.

Enabling SSH on IOS requires four things: a hostname, a domain name, an RSA key pair of at least 1024 bits, and a local user — then restrict the VTY lines to SSH only.

```
! minimum viable SSH setup
hostname CORE-SW1
ip domain-name oakacademy.local
crypto key generate rsa modulus 2048
username admin privilege 15 secret S0meStr0ngSecret
ip ssh version 2
line vty 0 15
  transport input ssh
  login local
```

1.18 What is a DoS attack, and how does a DDoS differ?

A denial-of-service attack makes a resource unavailable to legitimate users, either by exhausting something finite — bandwidth, connection state, CPU — or by triggering a crash. A SYN flood, which fills the server's half-open connection table, is the classic example.

A distributed denial-of-service does the same thing from thousands of compromised or reflecting hosts at once, which removes the option of simply blocking one source address and usually requires upstream scrubbing or provider-level filtering.

WHAT THEY'RE CHECKING

Some circulating question banks answer "DoS" as "Disk Operating System." In a network engineering interview, DoS means denial of service. Do not repeat that error.

The OSI & TCP/IP Models

Nobody asks about the OSI model because they care about the model. They ask because it is the shared vocabulary of troubleshooting, and because a candidate who can map a symptom to a layer can be handed a ticket on day one.

2.1 Why do we use a layered model at all?

Layering separates concerns so that each layer can be changed without rewriting the others. Fiber replaced copper at Layer 1 and TCP never noticed; HTTP/3 changed the transport underneath the same application semantics. It also gives vendors clean interoperability points, and gives engineers a structured way to isolate a fault instead of guessing.

2.2 List the seven OSI layers with the PDU and a protocol at each.

#	LAYER	PDU	EXAMPLES
7	Application	Data	HTTP, DNS, SMTP, FTP, SSH
6	Presentation	Data	Encoding, compression, TLS-style encryption
5	Session	Data	RPC, NetBIOS session setup and teardown
4	Transport	Segment / Datagram	TCP, UDP
3	Network	Packet	IPv4, IPv6, ICMP, OSPF
2	Data Link	Frame	Ethernet, 802.11, PPP, HDLC
1	Physical	Bit	Copper, fiber, RF, connectors, voltages

UDP's PDU is properly called a datagram, not a segment — a small distinction that lands well.

2.3 Explain encapsulation and decapsulation.

On the way down the stack, each layer prepends its own header to whatever it received from the layer above, treating that as opaque payload. Application data becomes a segment when TCP adds ports and sequence numbers, a packet when IP adds source and destination addresses, a frame when Ethernet adds MAC addresses and a trailing FCS, and finally a stream of bits on the wire.

The receiver reverses the process: each layer strips its own header, validates it, and hands the remainder upward. Ethernet is the only one of these that adds a trailer as well as a header.

2.4 Compare the OSI and TCP/IP models.

	OSI	TCP/IP
Layers	Seven	Four (Link, Internet, Transport, Application)
Origin	Designed as a reference model, then protocols	Protocols built first, model described afterward
Upper layers	Session and Presentation are separate	Both folded into Application
Use today	Teaching and troubleshooting vocabulary	What the internet actually runs on

The practical consequence: engineers speak OSI ("that's a Layer 2 problem") while operating TCP/IP. Both are correct in their own register.

2.5 What are the responsibilities of the Data Link layer?

Framing — marking where a frame starts and ends; physical addressing with MAC addresses; error *detection* using the frame check sequence; and media access control, deciding when a station may transmit.

Note the wording: Ethernet detects errors and discards bad frames. It does not correct them or ask for retransmission — that is left to TCP or the application.

WHAT THEY'RE CHECKING

The sublayer split is a frequent follow-up. LLC handles multiplexing to upper-layer protocols and optional flow control; the MAC sublayer handles framing, addressing, the FCS, and media access. The FCS belongs to MAC, not LLC.

2.6 What are the responsibilities of the Network layer?

Logical addressing, path determination, forwarding between different networks, and fragmentation when a packet is larger than the next link's MTU. ICMP lives here too, as the layer's own signaling and error-reporting mechanism.

2.7 What are the responsibilities of the Transport layer?

Segmentation and reassembly; multiplexing many conversations onto one host using port numbers; and, for TCP specifically, connection establishment, sequencing, acknowledgment, retransmission, flow control through the receive window, and congestion control.

UDP deliberately provides only multiplexing and a checksum, which is exactly why it is chosen for voice, video, and DNS.

2.8 Which layer do you troubleshoot first?

There are three defensible strategies, and naming them is the answer: **bottom-up** when you suspect a physical or new-installation problem, **top-down** when one application is broken but everything else works, and **divide-and-conquer** when you have a strong hypothesis and want to test at Layer 3 first with a ping.

In practice, start at Layer 1 for anything that has never worked, and start at the application for anything that worked yesterday and does not today.

2.9 A user reports "the internet is down." Diagnose it by layer.

- **L1** — is the port up? Link light, `show interface status`, cable, correct patch panel port.
- **L2** — is the port in the right VLAN, is it err-disabled, does the MAC appear in the table?
- **L3** — does the host have an IP and mask, is it 169.254.x.x, can it ping its gateway, then 8.8.8.8?
- **L4** — if the IP is reachable but the service is not, check ports and access lists.
- **L7** — if 8.8.8.8 pings but names fail, the problem is DNS. That is the single most common answer.

WHAT THEY'RE CHECKING

Structure, not speed. Say "I'd confirm the scope first — one user or the whole floor?" before you touch anything. Scoping is the habit that separates an engineer from a button-presser.

2.10 Which OSI layers do a switch, a router, and a firewall operate at?

A traditional switch is Layer 2; a multilayer switch also does Layer 3 routing between its SVIs. A router is Layer 3. A stateful firewall works at Layers 3 and 4 by tracking connection state, and a next-generation firewall inspects up to Layer 7 to identify applications and users regardless of port.

2.11 Where does TLS sit in the model?

Formally it is a session-layer construct presented as Presentation-layer services, running on top of TCP and beneath the application. Most engineers place it "between Layer 4 and Layer 7," and that answer is accepted. What matters is that you know TLS requires a reliable transport — which is why QUIC had to rebuild reliability over UDP to move HTTP/3 off TCP.

2.12 What is an MTU, and what happens when a packet exceeds it?

The MTU is the largest payload a Layer 2 frame can carry — 1500 bytes on standard Ethernet. If an IPv4 packet is larger and the Don't Fragment bit is clear, the router fragments it and the destination reassembles. If DF is set, the router drops it and returns ICMP type 3 code 4, "fragmentation needed," which is how Path MTU Discovery learns the limit.

This is why blocking all ICMP breaks large transfers over tunnels: PMTUD goes silent and you get a black hole where small packets and pings succeed but real traffic hangs.

RIP

RIP is nearly extinct in production, and interviewers know it. It survives on question lists because it is the cleanest way to test whether you understand distance-vector behavior and loop prevention — concepts that still matter in EIGRP and BGP.

3.1 What is RIP, and what are its key numbers?

RIP is a distance-vector interior gateway protocol whose only metric is hop count. It runs over UDP port 520, has an administrative distance of 120, supports a maximum of 15 hops, and treats 16 as infinity — unreachable. Version 2 sends updates to multicast 224.0.0.9; version 1 broadcasts to 255.255.255.255.

3.2 What does "distance vector" actually mean?

Each router advertises a distance (the metric) and a vector (the direction, meaning the next hop) for every prefix it knows. Critically, a distance-vector router never sees the topology — it only trusts what a neighbor tells it and adds its own cost. This is routing by rumor; and every loop-prevention mechanism in RIP exists to compensate for it.

Contrast that with link-state protocols like OSPF, where every router builds an identical map of the network and computes paths itself.

3.3 Name the RIP timers.

TIMER	DEFAULT	MEANING
Update	30 s	Interval between full periodic advertisements
Invalid	180 s	No update heard, so mark the route unreachable (metric 16)
Hold-down	180 s	Ignore worse information about that route while it stabilizes
Flush	240 s	Remove the route from the table entirely

Those numbers are why RIP convergence is measured in minutes. It is the protocol's fatal weakness and the reason EIGRP and OSPF replaced it.

3.4 What is the counting-to-infinity problem?

When a network goes down, two routers can keep advertising a path to it back and forth, each incrementing the hop count — 2, 3, 4 — believing the other has a valid route. Traffic loops between them until the metric finally reaches 16 and the prefix is declared unreachable. Capping the metric at 15 is what makes that termination possible at all.

3.5 Explain split horizon, route poisoning, poison reverse, and hold-down.

- **Split horizon** — never advertise a route back out the interface you learned it on.
- **Route poisoning** — when a network fails, advertise it with metric 16 rather than quietly withdrawing it, so neighbors learn of the failure explicitly.
- **Poison reverse** — on receiving a poisoned route, echo it back with metric 16 as an acknowledgment, overriding split horizon on purpose.
- **Hold-down** — after a route is poisoned, refuse worse information about it for a period so a stale update cannot resurrect the dead path.

Triggered updates complete the set: a topology change is sent immediately rather than waiting for the next 30-second cycle.

3.6 RIPv1 versus RIPv2 versus RIPng.

	RIPV1	RIPV2	RIPNG
Classful	Yes — no mask sent	No, sends the mask	No
VLSM / CIDR	Not supported	Supported	Supported
Delivery	Broadcast	Multicast 224.0.0.9	FF02::9, UDP 521
Authentication	None	Plaintext or MD5	Relies on IPsec

3.7 Configure RIPv2 correctly.

```
router rip
version 2
no auto-summary          ! keep VLSM intact
network 10.0.0.0          ! classful statement, enables matching interfaces
network 192.168.1.0
passive-interface Gi0/1  ! listen only, don't advertise to users
```

Two things are worth calling out. The `network` statement takes a classful address, not a mask, and it does not create the advertisement — it selects which interfaces participate. And `no auto-summary` is mandatory in any discontinuous design, because summarizing back to the classful boundary will silently break subnets split across a WAN.

3.8 What does passive-interface do, and why use it?

It suppresses routing updates out of an interface while still advertising that interface's subnet to other neighbors. You use it on any interface facing users or servers: there is no neighbor there to talk to, so sending updates only wastes bandwidth and hands an attacker a free topology map.

3.9 Why is RIP effectively obsolete?

Slow convergence driven by 30-second periodic updates and 180-second timers; a 15-hop diameter limit; a hop-count metric that treats a 10 Gbps link and a 1 Mbps link as equal; full-table updates rather than incremental ones; and no concept of hierarchy. EIGRP and OSPF solve every one of those.

WHAT THEY'RE CHECKING

The hop-count criticism is the strongest point you can make, because it shows you think about path quality rather than path existence. Offer it first.

EIGRP

EIGRP questions reward vocabulary precision. Successor, feasible successor, feasible distance, and reported distance are four different things, and interviewers use them as a fast test of whether your knowledge is structural or superficial.

4.1 What is EIGRP and how is it classified?

EIGRP is an advanced distance-vector — often called hybrid — protocol, originally Cisco proprietary and published as an informational RFC in 2013. It uses IP protocol number 88, multicasts to 224.0.0.10, and has an administrative distance of 90 for internal routes, 170 for external, and 5 for summary routes.

It is "advanced" because unlike RIP it maintains a topology table of alternate paths, sends incremental updates only on change, and converges through a computed algorithm rather than expiring timers.

4.2 Define successor, feasible successor, feasible distance, and reported distance.

- **Feasible distance (FD)** — your own best total metric to the destination.
- **Reported distance (RD)** — the metric your neighbor advertises, that is, the neighbor's own distance to the destination.
- **Successor** — the next hop on the best path. It goes in the routing table.
- **Feasible successor** — a guaranteed loop-free backup, kept in the topology table only, promoted instantly if the successor fails.

A path qualifies as a feasible successor when it satisfies the **feasibility condition**: the neighbor's reported distance must be strictly less than your current feasible distance. That test proves the neighbor is not routing back through you, which is what makes the backup loop-free without any recomputation.

WHAT THEY'RE CHECKING

Say the feasibility condition as an inequality, then say *why* it works — a neighbor closer to the destination than you cannot be sending traffic through you. Reciting the rule without the reason is the average answer.

4.3 What is DUAL?

The Diffusing Update Algorithm is EIGRP's path computation engine. When a successor is lost and a feasible successor exists, DUAL promotes it immediately — sub-second convergence with no recalculation. If no feasible successor exists, the route goes *active* and the router sends queries to its neighbors asking for an alternative, waiting for every reply before choosing a new path.

4.4 What is a stuck-in-active route?

A route stays active while the router waits for query replies. If a neighbor never replies — because of a broken link, high CPU, or a very deep query scope — the timer expires after three minutes and the router tears down the relationship with the unresponsive neighbor.

The cure is to limit query scope: summarization and stub routers stop queries from propagating across the whole network.

4.5 How is the EIGRP metric calculated?

The classic composite metric can use bandwidth, delay, load, reliability, and MTU, weighted by the K values. By default only K1 and K3 are set, so in practice the metric is a function of the **lowest bandwidth along the path plus the cumulative delay**. Load and reliability are off by default because they would make the metric fluctuate and destabilize the topology.

MTU is carried in the update but is a tie-breaker, not a metric component — a popular follow-up trap.

4.6 What are the EIGRP packet types?

Hello for neighbor discovery and keepalive; Update to advertise routes; Query to ask for an alternate path; Reply to answer a query; and Acknowledgment, which is a hello with no data used to confirm receipt of a reliable packet. Updates, queries, and replies are sent reliably via RTP; hellos and acks are not.

4.7 What must match for two routers to become EIGRP neighbors?

The autonomous system number, the K values, and — for classic EIGRP — the primary subnet on the shared link. Authentication must match if configured. Notably, hello and hold timers do *not* have to match, because each router advertises its own hold time; that is a difference from OSPF worth stating.

Defaults are hello 5 seconds and hold 15 on high-speed links, and hello 60 with hold 180 on low-speed multipoint interfaces.

4.8 What is an EIGRP stub router?

A stub advertises a limited set of routes — connected and summary by default — and tells its neighbors not to send it queries. In a hub-and-spoke design with hundreds of branches, configuring the spokes as stubs is the single most effective way to prevent stuck-in-active problems and keep query scope small.

4.9 Can EIGRP load-balance across unequal-cost paths?

Yes — and it is the only common IGP that can. The `variance` multiplier tells EIGRP to also install any feasible successor whose metric is within the multiplier times the best metric. A variance of 2 uses paths up to twice as costly as the best one. Only feasible successors are eligible, which is what keeps it loop-free.

4.10 Configure EIGRP and read its state.

```
router eigrp 100
  no auto-summary
  network 10.1.1.0 0.0.0.255      ! wildcard mask, not subnet mask
  network 172.16.0.0 0.0.255.255
  passive-interface default
  no passive-interface Gi0/0
  ! verification
  show ip eigrp neighbors
  show ip eigrp topology        ! FD / RD, successors and feasible successors
  show ip protocols
```

Note the wildcard mask in the network statement — the inverse of a subnet mask. Getting that backwards is one of the most common lab mistakes.

OSPF

The heaviest block in any routing interview, because OSPF is what most enterprises actually run. Expect at least one scenario question — a neighbor that won't come up, or an adjacency stuck in a particular state.

5.1 What is OSPF, and what are its key numbers?

OSPF is an open-standard link-state interior gateway protocol. It runs directly over IP protocol 89, uses multicast 224.0.0.5 for all routers and 224.0.0.6 for DR and BDR, has an administrative distance of 110, and computes shortest paths with Dijkstra's algorithm. Version 2 carries IPv4; version 3 carries IPv6.

Every router floods link-state advertisements describing its own links, assembles an identical link-state database, and then independently runs SPF against it. That shared map is what makes OSPF immune to routing-by-rumor loops.

5.2 How is OSPF cost calculated?

Cost equals a reference bandwidth divided by interface bandwidth, with a default reference of 100 Mbps. That gives FastEthernet a cost of 1, and every interface above 100 Mbps also a cost of 1 — so gigabit, ten-gigabit, and hundred-gigabit links appear identical to SPF.

The fix is `auto-cost reference-bandwidth` raised to at least 100000, applied consistently on every router in the domain, or setting `ip ospf cost` manually per interface.

WHAT THEY'RE CHECKING

Bringing up the reference-bandwidth problem unprompted is a strong senior signal. It is a real design defect that bites real networks, not a trivia item.

5.3 How does a router choose its OSPF router ID?

In order of preference: a manually configured `router-id`; otherwise the highest IP address on any up loopback interface; otherwise the highest IP address on any up physical interface. Always configure it manually — loopbacks are stable, but relying on election makes your router ID change the next time an interface flaps, and changing a router ID resets every adjacency.

5.4 List the OSPF adjacency states.

Down, Init, 2-Way, ExStart, Exchange, Loading, Full. On a broadcast segment, non-DR routers deliberately stop at 2-Way with each other and only reach Full with the DR and BDR — that is normal and not a fault.

The states are also a diagnostic tool. Stuck in **Init** means hellos are one-way, so suspect an access list or a unidirectional fiber. Stuck in **ExStart or Exchange** almost always means an **MTU mismatch** between the two interfaces.

5.5 What must match for an OSPF adjacency to form?

- Area ID, and both interfaces in the same subnet with the same mask
- Hello and dead intervals
- Authentication type and key
- MTU
- Stub area flag, and compatible network types
- Unique router IDs, and the interface must not be passive

Defaults are hello 10 and dead 40 seconds on broadcast and point-to-point links, and hello 30 with dead 120 on non-broadcast types.

5.6 How are the DR and BDR elected, and why do they exist?

They exist to control flooding. On a segment with n routers, full mesh adjacency would need $n(n-1)/2$ relationships; electing a designated router reduces that to $n-1$, with the DR relaying LSAs on behalf of everyone.

Election is by highest interface priority, then highest router ID as the tie-breaker. Priority 0 means never eligible. The election is **non-preemptive** — a better router joining later does not take over until the current DR fails, which surprises candidates who assume it behaves like an STP root election.

DR and BDR are elected only on broadcast and non-broadcast multi-access networks. Point-to-point links have no DR.

5.7 Why does OSPF use areas, and what is Area 0?

Areas bound the scope of SPF recalculation and LSA flooding. A topology change inside one area forces a recalculation only in that area; other areas see, at most, a summary update. Without areas, every flap in a large network makes every router rerun Dijkstra.

Area 0 is the backbone, and every other area must connect to it — either physically or through a virtual link across a transit area. That rule is what guarantees a loop-free two-level hierarchy.

5.8 Define internal router, ABR, ASBR, and backbone router.

An **internal router** has all interfaces in one area. An **ABR** (area border router) has interfaces in two or more areas, one of which must be Area 0, and it generates summary LSAs between them. An **ASBR** redistributes routes from outside OSPF — static, BGP, another IGP — and generates external LSAs. A **backbone router** is any router with an interface in Area 0.

5.9 Name the LSA types you need to know.

TYPE	NAME	GENERATED BY	SCOPE
1	Router	Every router	Within its area
2	Network	The DR	Within its area
3	Summary	ABR	Between areas
4	ASBR summary	ABR	Path to the ASBR
5	External	ASBR	Whole domain except stubs
7	NSSA external	ASBR in an NSSA	Within the NSSA, converted to type 5 at the ABR

5.10 What is a stub area, and what are the variants?

A **stub area** blocks type 5 external LSAs and replaces them with a default route, shrinking the database on small routers. A **totally stubby area** — a Cisco feature — also blocks type 3 inter-area summaries, leaving only a default. An **NSSA** is a stub that is nonetheless allowed to have its own ASBR, using type 7 LSAs that the ABR translates into type 5 on the way out.

Area 0 can never be a stub, since all external and inter-area traffic transits it.

5.11 What is a virtual link?

A logical tunnel through a non-backbone transit area that connects an orphaned area to Area 0, used when a merger or a topology change leaves an area with no physical path to the backbone. It works, but treat it as a temporary repair: it complicates troubleshooting and cannot cross a stub area.

5.12 What are the OSPF network types?

Broadcast (Ethernet — DR/BDR elected), point-to-point (serial and subinterfaces — no DR, fastest adjacency), non-broadcast NBMA, point-to-multipoint, and point-to-multipoint non-broadcast. The type controls whether a DR is elected and whether neighbors are discovered automatically or must be configured statically.

5.13 Configure single-area OSPF and verify it.

```
router ospf 1
  router-id 1.1.1.1
  auto-cost reference-bandwidth 100000
  passive-interface default
  no passive-interface Gi0/0
  network 10.1.1.0 0.0.0.255 area 0
! or, per interface, which is now preferred:
interface Gi0/0
  ip ospf 1 area 0
  ip ospf priority 100
! verification order that reads well in an interview
show ip ospf neighbor      ! is the adjacency Full?
show ip ospf interface brief ! right area, cost, timers?
show ip route ospf         ! did the prefix install?
show ip ospf database
```

The OSPF process ID is locally significant only — two routers with process 1 and process 65 will still peer, as long as the area and the parameters above agree.

Access Control Lists

ACLs are where networking meets security, which is why they come up in almost every interview for a security-adjacent role. Expect to be handed a requirement and asked to write the lines.

6.1 What is an ACL and where can it be applied?

An access control list is an ordered set of permit and deny statements matched against traffic. Its two jobs are filtering — applied to an interface with `ip access-group` — and classification, where it identifies traffic for NAT, route maps, QoS, or VPN interesting-traffic definitions. Only one ACL can be applied per interface, per direction, per protocol.

6.2 Standard versus extended ACLs.

	STANDARD	EXTENDED
Numbers	1–99, 1300–1999	100–199, 2000–2699
Matches on	Source IP only	Source, destination, protocol, ports, flags
Place it	Close to the destination	Close to the source

The placement logic follows from the matching capability. A standard ACL near the source would block that host from reaching *everything*, not just the one destination you meant. An extended ACL is precise enough to drop unwanted traffic at the first hop and save bandwidth across the network.

6.3 What happens to traffic that matches nothing?

It is dropped. Every ACL ends with an invisible `deny ip any any`. This means an ACL containing only deny statements blocks everything, and it means that the moment you apply any ACL you must also explicitly permit the traffic you still need — including management access, DHCP, and routing protocol adjacencies.

WHAT THEY'RE CHECKING

The follow-up is usually "so how would you see what's being dropped?" Add an explicit `deny ip any any log` at the end. It changes nothing functionally but gives you counters and syslog visibility.

6.4 Why does the order of ACL entries matter?

Processing is strictly top-down and stops at the first match. A broad permit above a specific deny makes the deny unreachable — the classic misconfiguration. Write from most specific to most general, and always verify with `show access-lists`, where the hit counters tell you which lines traffic is actually matching.

6.5 Explain wildcard masks.

A wildcard mask is the inverse of a subnet mask: a 0 bit means "this bit must match," a 1 bit means "ignore this bit." So 0.0.0.255 matches a whole /24, 0.0.0.0 matches exactly one host, and 255.255.255.255 matches anything. The keywords `host` and `any` are shorthand for the first and last of those.

To derive one, subtract each subnet mask octet from 255: a /26 mask of 255.255.255.192 becomes a wildcard of 0.0.0.63.

6.6 Write an extended ACL for a real requirement.

Requirement: the 10.10.10.0/24 user subnet may reach the web server 192.168.50.10 on HTTPS only; everything else from that subnet toward the server farm is denied; all other traffic passes.

```
ip access-list extended USERS-T0-DMZ
  permit tcp 10.10.10.0 0.0.0.255 host 192.168.50.10 eq 443
  deny ip 10.10.10.0 0.0.0.255 192.168.50.0 0.0.0.255
  permit ip any any
  deny ip any any log          ! unreachable, but keeps counters honest
interface Gi0/1
  ip access-group USERS-T0-DMZ in
```

Named ACLs are preferred over numbered ones because they document intent and support sequence numbers, so you can insert or delete a single line instead of rewriting the list.

6.7 Inbound or outbound — how do you choose the direction?

Direction is always relative to the interface, not to the network. Inbound filtering is more efficient because the router drops the packet before the routing lookup, and it is the right choice for anti-spoofing at an edge. Outbound is right when you want traffic from several inbound interfaces filtered at one exit point.

One subtlety worth mentioning: an outbound ACL does not filter traffic generated by the router itself.

6.8 How do you protect management access with an ACL?

With `access-class` on the VTY lines rather than `ip access-group` on an interface — that way the restriction applies regardless of which interface the session arrives on.

```
ip access-list standard MGMT-HOSTS
  permit 10.99.0.0 0.0.0.255
line vty 0 15
  access-class MGMT-HOSTS in
  transport input ssh
  exec-timeout 10 0
```

6.9 What does the established keyword do?

It matches TCP segments whose ACK or RST flag is set, meaning the segment belongs to a conversation that was initiated in the opposite direction. It lets internal users receive replies from the internet without allowing outsiders to open new inbound connections.

It is a rough approximation of statefulness — the router is inspecting a flag, not tracking a session — so a real design uses a stateful firewall or Cisco's zone-based policy firewall instead.

6.10 You applied an ACL and now the routing protocol has dropped. Why?

Because the implicit deny is also denying the protocol's own traffic. OSPF is IP protocol 89 to 224.0.0.5 and 224.0.0.6; EIGRP is protocol 88 to 224.0.0.10. Neither is TCP or UDP, so a rule permitting "tcp any any" does not cover them. The same trap catches DHCP (UDP 67 and 68 to a broadcast address) and ICMP.

WHAT THEY'RE CHECKING

Whether you think about control-plane traffic, not just user traffic. Mentioning this unprompted signals real hands-on time.

NAT & PAT

Everyone can define NAT. Fewer candidates can name the four address types correctly, and fewer still can say what NAT breaks. Both of those are the actual test.

7.1 What is NAT and why do we use it?

Network Address Translation rewrites the IP addresses — and with PAT, the port numbers — in packet headers as they cross a boundary. The original driver was IPv4 exhaustion: thousands of private hosts can share a handful of public addresses. Secondary benefits are topology hiding and the ability to merge networks with overlapping address space.

Be clear that NAT is not a security control. It hides internal structure as a side effect, but a stateful firewall is what enforces policy.

7.2 Define inside local, inside global, outside local, and outside global.

TERM	MEANING	EXAMPLE
Inside local	The internal host's real private address	10.1.1.10
Inside global	How that host appears to the outside after translation	203.0.113.5
Outside global	The external host's real public address	8.8.8.8
Outside local	How the external host appears to the inside	8.8.8.8, unless you also translate the destination

The pattern that makes this stick: *inside* and *outside* tell you whose address it is; *local* and *global* tell you which side of the router you are standing on when you look at it.

7.3 Static NAT, dynamic NAT, and PAT — when do you use each?

- **Static NAT** — a permanent one-to-one mapping. Required for any server that must be reachable from outside, since the mapping exists before any traffic flows.
- **Dynamic NAT** — one-to-one from a pool, assigned on demand. When the pool is exhausted, further hosts simply fail.
- **PAT / NAT overload** — many-to-one, distinguished by source port. This is what every home and branch router does, and it supports roughly tens of thousands of concurrent sessions per public address.

7.4 Configure PAT.

```
ip access-list standard NAT-USERS
  permit 10.1.1.0 0.0.0.255
interface Gi0/0
  ip nat inside
interface Gi0/1
  ip nat outside
ip nat inside source list NAT-USERS interface Gi0/1 overload
! verification
show ip nat translations
show ip nat statistics
clear ip nat translation *
```

The `overload` keyword is what turns this from dynamic NAT into PAT. Forgetting the `ip nat inside` and `ip nat outside` designations is the most common reason a correct-looking NAT configuration does nothing at all.

7.5 How do you publish an internal server on a single public IP?

Static PAT, also called port forwarding — you map a single port rather than the whole address:

```
ip nat inside source static tcp 10.1.1.20 443 interface Gi0/1 443
```

That leaves the public address free to keep serving outbound PAT for everyone else.

7.6 What does NAT break?

- **End-to-end addressing** — no unsolicited inbound connections, which is why peer-to-peer and VoIP need STUN, TURN, or ICE.
- **Protocols that embed addresses in the payload** — active FTP, SIP, and some legacy applications need an application-layer gateway to rewrite the payload too.
- **IPsec AH**, which authenticates the IP header itself. ESP survives via NAT-Traversal, which encapsulates it in UDP 4500.
- **Attribution** — hundreds of users behind one address make logs and incident response harder, which is exactly why carrier-grade NAT logging is a compliance topic.

WHAT THEY'RE CHECKING

For a security-track role this is the question that matters most in the whole NAT block. Lead with the attribution and logging point.

7.7 In what order does a router process NAT and routing?

For inside-to-outside traffic, the router routes first and then translates. For outside-to-inside, it translates first and then routes. That ordering explains a classic symptom: an ACL applied inbound on the outside interface sees the *untranslated* public destination address, so your rules must be written against the public address, not the private one.

7.8 Does NAT solve IPv4 exhaustion?

It postponed it — arguably by two decades — at the cost of breaking the end-to-end model and adding stateful complexity to the middle of the network. IPv6 is the actual solution, with a 128-bit address space and no need for translation. Most enterprises now run dual-stack, so the honest answer is that you need to be fluent in both.

7.9 A user behind PAT cannot reach the internet. How do you troubleshoot?

Work the chain in order: is there a route to the destination, including a default route out; are the inside and outside interfaces designated; does the ACL or pool actually match the user's source subnet; is the pool exhausted; and does `show ip nat translations` show an entry being created when the user generates traffic? No entry means the match is failing; an entry with no return traffic means the problem is upstream or in an ACL.

7.10 What is the difference between NAT and a proxy?

NAT operates at Layer 3 and 4 and rewrites headers while leaving the connection logically end-to-end. A proxy terminates the client's connection at the application layer and opens a separate one of its own, which lets it cache, authenticate users, inspect content, and enforce URL policy. NAT is transparent plumbing; a proxy is a deliberate application-layer intermediary.

DHCP

DHCP is the most common source of "the network is down" tickets, so interviewers use it to test troubleshooting instincts. The security questions attached to it — starvation, rogue servers, snooping — are increasingly the real focus.

8.1 Walk me through the DHCP process.

Four messages, remembered as **DORA**:

- **Discover** — the client broadcasts from 0.0.0.0 to 255.255.255.255, source port 68 to destination 67.
- **Offer** — one or more servers propose an address, mask, gateway, DNS, and lease time.
- **Request** — the client broadcasts its acceptance of one specific offer, which also tells the other servers to release their reservations.
- **Acknowledge** — the chosen server confirms and commits the lease.

The Request is broadcast rather than unicast precisely so the losing servers hear it. That detail is a good discriminator.

8.2 Which ports does DHCP use, and why does it work before the client has an IP?

UDP 67 on the server and UDP 68 on the client. It works because the exchange happens entirely at Layer 2 broadcast level: the client has no address, so it uses 0.0.0.0 as the source and the all-ones broadcast as the destination, and the switch floods it within the VLAN. Everything about DHCP's design follows from the fact that the client cannot yet be routed to.

8.3 The DHCP server is in another subnet. How does the client reach it?

Through a DHCP relay agent. You configure `ip helper-address` on the router interface facing the clients; the router converts the broadcast into a unicast toward the server and inserts its own interface address in the giaddr field so the server knows which scope to serve from.

```
interface Vlan20
 ip address 10.20.0.1 255.255.255.0
 ip helper-address 10.99.0.10
```

8.4 What is a DHCP scope, and what would you exclude from it?

A scope is the range of addresses a server may lease on a given subnet, together with the options it hands out. You exclude anything statically assigned: the gateway, switch and router management addresses, printers, servers, and access points. Reservations bind a specific address to a MAC for devices that need a predictable address but should still be centrally managed.

8.5 How does lease renewal work?

At 50 percent of the lease (timer T1) the client unicasts a Request directly to its current server to extend the lease. If that fails, at 87.5 percent (T2) it broadcasts, willing to accept any server. If the lease expires entirely, it releases the address and starts over with Discover.

Short leases suit guest and wireless networks with high turnover; long leases reduce server load on stable wired segments.

8.6 A client has a 169.254.x.x address. What does that tell you?

That the client asked for DHCP and got no answer, so it self-assigned a link-local APIPA address. It can talk to other APIPA hosts on the same segment and nothing else. The cause is upstream: no server, an exhausted scope, a missing helper-address, the port in the wrong VLAN, or DHCP snooping dropping the offer on an untrusted port.

WHAT THEY'RE CHECKING

Scope the fault immediately. One machine points at that port or NIC; a whole VLAN points at the scope, the relay, or the server.

8.7 Configure a router as a DHCP server.

```
ip dhcp excluded-address 10.20.0.1 10.20.0.20
ip dhcp pool VLAN20-USERS
  network 10.20.0.0 255.255.255.0
  default-router 10.20.0.1
  dns-server 10.99.0.53 10.99.0.54
  domain-name oakacademy.local
  lease 0 8 0 ! 8 hours
! verification
show ip dhcp binding
show ip dhcp pool
show ip dhcp conflict
```

8.8 What is a DHCP starvation attack, and what is a rogue DHCP server?

Starvation floods the server with Discover messages using spoofed MAC addresses until the pool is empty, denying service to real clients. A rogue server then answers the abandoned clients with its own offer, handing out an attacker-controlled default gateway or DNS server — a clean man-in-the-middle setup. The two are usually run together.

8.9 How does DHCP snooping stop that?

DHCP snooping classifies every port as trusted or untrusted. Server-side and uplink ports are trusted; access ports are untrusted. On an untrusted port the switch drops server-originated messages — Offer and Ack — so a rogue server on a user port cannot answer anyone. Rate limiting on untrusted ports blunts starvation, and the switch builds a binding table of legitimate IP-to-MAC-to-port mappings.

That binding table is then the foundation for Dynamic ARP Inspection and IP Source Guard, which is why snooping is the first Layer 2 hardening step you enable.

```
ip dhcp snooping
ip dhcp snooping vlan 20
interface Gi1/0/24
  description uplink to distribution
  ip dhcp snooping trust
interface range Gi1/0/1-23
  ip dhcp snooping limit rate 15
```

8.10 How does address assignment differ in IPv6?

IPv6 hosts can self-configure with SLAAC, deriving an address from the prefix in a router advertisement without any server at all. DHCPv6 still exists for cases where you need centralized control and options — stateful DHCPv6 assigns the address, while stateless DHCPv6 lets SLAAC do the addressing and supplies only options such as DNS. There is no broadcast, so the client uses the multicast address ff02::1:2 on UDP 546 and 547.

TCP & UDP

The transport layer is where networking and security interviews overlap most: handshakes, state, and ports are the vocabulary of both firewall rules and attack detection.

9.1 Compare TCP and UDP.

	TCP	UDP
Connection	Connection-oriented, three-way handshake	Connectionless, just sends
Reliability	Acknowledgment, sequencing, retransmission	None — the application handles it or accepts loss
Ordering	Guaranteed by sequence numbers	Not guaranteed
Flow control	Sliding window	None
Header	20 bytes minimum, up to 60 with options	8 bytes, fixed
Typical use	HTTP, SSH, SMTP, file transfer	DNS queries, VoIP, video, TFTP, SNMP, syslog

The framing that impresses: TCP trades latency for reliability, UDP trades reliability for latency. Neither is better — you pick based on whether a late packet is still useful. In a voice call it is not, so retransmitting it would make things worse.

9.2 Describe the three-way handshake.

The client sends **SYN** with its initial sequence number. The server replies **SYN-ACK**, acknowledging the client's ISN and offering its own. The client sends **ACK**. Both sides now have synchronized sequence numbers and agreed window sizes, and the connection is ESTABLISHED.

Two independent, randomized initial sequence numbers are what make the handshake meaningful — they prove each side can receive from the other, which defeats naive source-address spoofing.

9.3 How does a TCP connection close?

With a four-way exchange, because each direction is closed independently: FIN, ACK, then FIN, ACK from the other side. The side that closes first enters TIME_WAIT for twice the maximum segment lifetime, so that any straggling segments die out before the same port pair can be reused.

A RST is the abrupt alternative — an immediate teardown, sent when a port is closed or a socket is aborted.

9.4 Name the TCP flags and what each one signals.

SYN opens a connection; ACK acknowledges data; FIN requests a graceful close; RST forces an abrupt reset; PSH asks the receiver to deliver buffered data to the application immediately; URG marks urgent data. In practice, security work revolves around SYN, ACK, FIN, and RST — a SYN with no matching ACK is a half-open connection, and a flood of them is the signature of a SYN flood or a stealth scan.

9.5 What is a window, and what does sliding window mean?

The window is the number of bytes a sender may have outstanding before it must stop and wait for an acknowledgment — advertised by the *receiver*, representing its available buffer. It slides forward as acknowledgments arrive, so transmission is continuous rather than stop-and-wait.

A window of zero is a legitimate signal meaning "stop sending, my buffer is full." Window scaling extends the 16-bit field so that high-bandwidth, high-latency paths can be filled.

WHAT THEY'RE CHECKING

Whether you know the window is set by the receiver, not the sender. Getting that backwards is a common and very visible error.

9.6 Flow control or congestion control — what is the difference?

Flow control protects the *receiver* from being overrun, and works through the advertised window. Congestion control protects the *network* from being overrun, and works through the sender's own congestion window using slow start, congestion avoidance, fast retransmit, and fast recovery. One responds to a buffer, the other to packet loss as an implicit congestion signal.

9.7 What is MSS, and how does it relate to MTU?

The maximum segment size is the largest payload TCP will place in one segment, negotiated in the SYN. On standard Ethernet it is the 1500-byte MTU minus 20 bytes of IP header and 20 of TCP header, so 1460. Tunnels reduce it further, which is why GRE or IPsec overlays often need `ip tcp adjust-mss 1360` to avoid fragmentation and PMTUD black holes.

9.8 Explain the port number ranges.

0–1023 are well-known and assigned by IANA; 1024–49151 are registered; 49152–65535 are dynamic or ephemeral, used as the client's source port. A conversation is identified by the full five-tuple — source IP, source port, destination IP, destination port, and protocol — which is how one server can hold thousands of simultaneous connections on port 443.

9.9 Give me the port numbers you should know cold.

SERVICE	PORT	SERVICE	PORT
FTP data / control	20 / 21 TCP	HTTP	80 TCP
SSH	22 TCP	POP3	110 TCP
Telnet	23 TCP	NTP	123 UDP
SMTP	25 TCP	IMAP	143 TCP
DNS	53 UDP and TCP	SNMP / traps	161 / 162 UDP
DHCP server / client	67 / 68 UDP	LDAP	389 TCP
TFTP	69 UDP	HTTPS	443 TCP
SMB	445 TCP	Syslog	514 UDP
RDP	3389 TCP	Radius	1812 / 1813 UDP

9.10 Which service uses both TCP and UDP, and why?

DNS. Ordinary queries use UDP 53 because a single small request and reply does not justify a handshake. TCP 53 is used for zone transfers and for any response too large for a single datagram, since TCP provides the reliability and segmentation that a large transfer needs. DNS over TLS and DNS over HTTPS have since moved encrypted resolution to TCP 853 and 443.

9.11 How does a SYN flood work, and how do you defend against it?

The attacker sends a flood of SYNs from spoofed sources and never completes the handshake. Each one consumes an entry in the server's half-open connection table until legitimate clients can no longer be accepted. Defenses are SYN cookies, which let the server avoid allocating state until the final ACK arrives; shortened SYN timeouts; connection rate limiting at the edge; and upstream scrubbing for volumetric attacks.

9.12 Which TCP states matter in practice?

LISTEN means a service is waiting for connections. SYN_SENT and SYN_RECEIVED are mid-handshake. ESTABLISHED is an active conversation. FIN_WAIT and CLOSE_WAIT are stages of teardown — a pile-up of CLOSE_WAIT usually indicates an application that is not closing its sockets. TIME_WAIT is the deliberate cooling-off period; thousands of them are normal on a busy server, not a fault.

The IPv4 Header

Header questions are a packet-capture literacy test. If you have ever opened Wireshark with intent, these are easy; if you have only read about it, they are not.

10.1 How large is the IPv4 header, and what does it contain?

Twenty bytes without options, up to sixty with them. The fields are version, IHL (header length), DSCP and ECN — the field originally called Type of Service — total length, identification, flags, fragment offset, TTL, protocol, header checksum, source address, destination address, and optional options with padding.

10.2 What is TTL for?

Time to Live is a hop counter, not a clock. Every router decrements it by one, and any router that decrements it to zero discards the packet and returns ICMP time-exceeded to the source. It is the mechanism that guarantees a routing loop consumes finite resources rather than looping forever.

It is also a fingerprint: the initial value differs by operating system — 64 on Linux and macOS, 128 on Windows — so the received TTL tells you both the sender's platform and roughly how many hops away it is.

10.3 How does traceroute use TTL?

It sends probes with TTL 1, then 2, then 3. Each router one hop further out is forced to discard the probe and reply with ICMP time-exceeded, revealing its own address. When the destination is finally reached, it answers differently — a port-unreachable or an echo reply — which signals the end of the path.

Implementations differ: Windows `tracert` uses ICMP echo requests, while the classic Unix `traceroute` sends UDP probes to high ports starting around 33434, with an ICMP mode available as an option. That difference explains why one of them completes through a firewall and the other does not.

WHAT THEY'RE CHECKING

Also know why traceroute shows asterisks: a hop that is configured not to send ICMP time-exceeded is invisible even though traffic passes through it perfectly. Missing hops are not proof of a break.

10.4 What does the Protocol field do? Give some values.

It tells the receiving stack which upper-layer protocol to hand the payload to. Values worth memorizing: 1 is ICMP, 6 is TCP, 17 is UDP, 47 is GRE, 50 is ESP, 51 is AH, 88 is EIGRP, and 89 is OSPF.

Those last two are the reason a firewall rule permitting only TCP and UDP silently kills your routing adjacencies.

10.5 Explain the identification, flags, and fragment offset fields.

All three exist to support fragmentation. Identification tags every fragment of the same original packet with a common value; the flags field carries Don't Fragment and More Fragments; the offset states where this fragment belongs in the reassembled whole. Only the destination reassembles — routers never do.

Fragmentation is also an evasion technique, which is why intrusion prevention systems reassemble traffic before inspecting it.

10.6 Why is the header checksum recomputed at every hop?

Because the header changes at every hop — TTL is decremented, and the checksum covers the header only, so it must be recalculated. It does not cover the payload; TCP and UDP have their own checksums for that. IPv6 removed the header checksum entirely, on the reasoning that Layer 2 and Layer 4 already detect corruption and the per-hop recomputation was pure overhead.

10.7 What is DSCP?

Differentiated Services Code Point — six bits in the former Type of Service byte used to mark a packet's QoS class so that every downstream device can queue it consistently. EF (expedited forwarding, decimal 46) is the standard marking for voice; AF41 is common for video; the default is best effort.

The design principle: mark as close to the source as possible and trust the marking only from devices you control, re-marking anything arriving from an untrusted edge.

10.8 How does the IPv6 header differ?

It is a fixed 40 bytes with only eight fields. The checksum is gone, fragmentation fields are gone — routers never fragment IPv6, the source must handle it — options moved to optional extension headers, and TTL was renamed Hop Limit. The result is a header that is larger in bytes but dramatically cheaper for a router to process.

10.9 Which header fields change as a packet crosses a router?

TTL decrements and the header checksum is recalculated. If NAT is involved, the source or destination address changes too. Everything else — including the source and destination addresses on a normally routed packet — stays intact end to end. The Layer 2 header, by contrast, is discarded and rebuilt at every hop.

ICMP

ICMP looks like a small topic and then turns into a security discussion. The best answer to "should you block ICMP" is the one that shows you understand what breaks if you do.

11.1 What is ICMP and which layer does it belong to?

The Internet Control Message Protocol is IP's own signaling and error-reporting mechanism, IP protocol number 1. It sits at Layer 3, is encapsulated directly in IP, and has no port numbers — which is why firewall rules for ICMP are written by type and code rather than by port.

ICMP reports problems; it never fixes them. It delivers no data on behalf of applications.

11.2 Name the ICMP types and codes you should know.

TYPE	MEANING	NOTABLE CODES
0	Echo reply	The answer to a ping
3	Destination unreachable	0 network, 1 host, 3 port, 4 fragmentation needed, 13 administratively prohibited
5	Redirect	Use a better gateway for this destination
8	Echo request	The ping itself
11	Time exceeded	0 TTL expired in transit — how traceroute works

11.3 What does ping actually prove?

That Layer 3 reachability exists in both directions and that the destination's stack is responding. It proves nothing about whether an application is running, whether a port is open, or whether the path is fast enough to be usable.

A failed ping proves even less, since many hosts and firewalls drop echo requests by policy. "It doesn't ping" is a data point, not a diagnosis.

11.4 Interpret the output characters of a Cisco ping.

A period is a timeout; an exclamation mark is a successful reply; **U** means a router returned destination-unreachable; **Q** is source quench; **M** means fragmentation was needed but DF was set; and **?** is an unknown packet type. The distinction that matters: a period means silence, while **U** means something in the path answered and told you it could not deliver — which is far more useful information.

11.5 What is an extended ping and when would you use it?

In privileged mode, typing `ping` with no arguments prompts for options: repeat count, datagram size, source interface or address, and the DF bit. The two high-value uses are sourcing the ping from a specific interface — to test return routing and ACLs from the right address — and setting DF with increasing sizes to discover the real path MTU.

11.6 What is an ICMP redirect?

A message from a router telling a host that a different gateway on the same subnet is a better next hop for a given destination. It is legitimate on a segment with two routers, but it is also an attack primitive — a forged redirect can steer a victim's traffic through an attacker — so it is commonly disabled with `no ip redirects` on user-facing interfaces.

11.7 How has ICMP been abused?

Ping sweeps for reconnaissance; the Smurf attack, which amplified echo requests off a directed broadcast address; ping of death, using malformed oversized fragments; and ICMP tunneling, where data or command-and-control traffic is hidden inside echo payloads to bypass egress filtering. The last one is the reason security teams alert on unusually large or high-volume ICMP.

11.8 Should you block ICMP at the perimeter?

Not wholesale. Blocking echo requests inbound from the internet is reasonable. Blocking ICMP entirely breaks Path MTU Discovery — you need type 3 code 4 — and breaks traceroute, which needs type 11. The result is the classic black hole where small packets and pings work while HTTPS and VPN sessions hang.

WHAT THEY'RE CHECKING

Nuance. "Block all ICMP, it's a security risk" is the answer of someone who has never debugged a tunnel. Name the two types you must permit and you have answered better than most candidates.

ARP

ARP is a small protocol with no authentication whatsoever, which makes it both a fundamental mechanism and one of the most reliable attack surfaces on a local network. Interviewers use it to bridge from networking into security.

12.1 What does ARP do?

It resolves a known IPv4 address to the unknown MAC address on the same broadcast domain, so that the sender can build a frame. The request is a Layer 2 broadcast — "who has 10.1.1.1?" — and the reply is a unicast from the owner. The result is cached with a timeout, typically four hours on Cisco devices and much shorter on hosts.

12.2 A host pings a device in another subnet. Whose MAC does it ARP for?

The default gateway's. The host compares the destination against its own subnet, sees that it is remote, and therefore needs the MAC of the router — not of the far-end host, which it can never learn directly. The frame carries the gateway's MAC as destination but the remote host's IP as destination.

WHAT THEY'RE CHECKING

This is the single best test of whether a candidate has truly internalized the two-address model. Answer it crisply and most of the fundamentals block is settled.

12.3 Does ARP cross a router?

No. ARP is a Layer 2 broadcast and routers do not forward broadcasts, so ARP is confined to a single broadcast domain — the definition of the "local segment" that matters. Every router hop performs its own independent ARP resolution for its next hop.

12.4 What is a gratuitous ARP?

An unsolicited ARP reply announcing an address binding that nobody asked for. Legitimate uses are duplicate-address detection at boot, updating neighbors after a NIC replacement, and failover — an HSRP standby or a clustered server sends one so switches and hosts immediately learn the new location of the virtual address.

The same message is the vehicle for ARP poisoning, since nothing in ARP verifies who is entitled to claim an address.

12.5 Explain ARP spoofing and how to stop it.

The attacker sends forged ARP replies claiming the gateway's IP belongs to the attacker's MAC. Victims update their caches and send all off-subnet traffic to the attacker, who forwards it on — a transparent man in the middle for sniffing or modification.

The mitigation is **Dynamic ARP Inspection**, which validates ARP packets on untrusted ports against the DHCP snooping binding table and drops any that do not match. That is why DHCP snooping must be enabled first. Static ARP entries and port security help at the margins, and encryption limits the damage of a successful attack.

12.6 What is proxy ARP?

A router answering ARP requests on behalf of a host in a different subnet, using its own MAC. It was designed to let hosts with a wrong or missing subnet mask still communicate. It hides configuration errors, enlarges ARP tables, and can be abused, so modern practice is to disable it with `no ip proxy-arp`.

12.7 What is RARP and what replaced it?

Reverse ARP let a diskless workstation ask "I know my MAC, what is my IP?" It was superseded by BOOTP and then by DHCP, which return not only an address but a mask, gateway, DNS servers, and arbitrary options. RARP is a historical answer only.

12.8 How does IPv6 handle address resolution?

There is no ARP. The Neighbor Discovery Protocol, carried over ICMPv6, uses Neighbor Solicitation and Neighbor Advertisement messages sent to a solicited-node multicast address rather than a broadcast — so only the likely owner is disturbed. NDP also covers router discovery, prefix advertisement for SLAAC, and duplicate address detection.

12.9 Which commands do you use to inspect ARP?

```
show ip arp           ! router or switch ARP cache
show arp              !
clear arp-cache       ! force re-resolution
show ip arp inspection ! DAI status and drop counters
arp -a                ! Windows / Linux host side
```

A useful field check: two different IPs mapping to the same MAC on a user VLAN is either a legitimate router or a poisoning attempt in progress.

SNMP & Syslog

Monitoring questions signal that the role involves operations, not just build work. Being able to explain trap versus poll, and severity levels, marks you as someone who has been on call.

13.1 What is SNMP and what are its components?

The Simple Network Management Protocol lets a central system monitor and configure network devices. It has three parts: the **manager** (the NMS), the **agent** running on each device, and the **MIB** — a hierarchical database of managed objects, each identified by an OID. The agent listens on UDP 161; the manager receives traps and informs on UDP 162.

13.2 Name the SNMP operations.

Get retrieves one object; GetNext walks to the following object in the tree; GetBulk (added in v2c) retrieves many at once efficiently; Set writes a value and therefore changes configuration; Trap is an unsolicited alert from the agent; and Inform is a trap that requires acknowledgment and will be retransmitted if unacknowledged.

13.3 Polling or traps — which do you use?

Both, for different jobs. Polling gives you the time-series data you need for trends and capacity planning, but it only sees the state at the moment it asks. Traps deliver events the instant they happen, so you learn about an interface flap in seconds rather than at the next poll interval. A monitoring system without traps is always late; one without polling has no history.

13.4 Compare SNMP v1, v2c, and v3.

VERSION	SECURITY	NOTES
v1	Community string in cleartext	Obsolete, 32-bit counters
v2c	Community string in cleartext	GetBulk, 64-bit counters, still widely deployed
v3	Authentication, integrity, and optional encryption	The only version acceptable on a modern network

Version 3 offers three security levels: noAuthNoPriv, authNoPriv, and authPriv. Only authPriv both verifies the sender and encrypts the payload.

13.5 Why is a community string not a password?

Because it is transmitted in cleartext with no hashing or challenge, so a single capture reveals it, and it identifies a group rather than a user, giving no accountability. A read-write community string is effectively an unauthenticated configuration channel — which is why "public" and "private" left at their defaults are perennial audit findings.

Practical hardening: prefer v3, otherwise use read-only strings, restrict access with an ACL, and never expose UDP 161 to an untrusted network.

13.6 Configure SNMP and syslog on IOS.

```
! SNMPv3 - authenticated and encrypted
snmp-server group MONITOR v3 priv read READONLY access 20
snmp-server user nms MONITOR v3 auth sha AuthPass123 priv aes 128 PrivPass456
snmp-server host 10.99.0.20 version 3 priv nms
! syslog to a central collector
logging host 10.99.0.30
logging trap informational
logging source-interface Loopback0
service timestamps log datetime msec localtime show-timezone
ntp server 10.99.0.1
```

13.7 List the syslog severity levels.

LEVEL	NAME	LEVEL	NAME
0	Emergency — system unusable	4	Warning
1	Alert — immediate action	5	Notification
2	Critical	6	Informational — default on IOS
3	Error	7	Debugging

Lower is more severe, and configuring a level sends that level *and everything more severe*. Level 7 on a production device will flood your collector and can spike CPU — a real operational trap.

13.8 Why does NTP matter for logging?

Because correlation is impossible without a common clock. If three devices disagree by minutes, you cannot reconstruct the order of events in an incident, and timestamps lose evidentiary value. Every device should sync to the same NTP source, log in a consistent time zone or in UTC, and include the date and millisecond precision in its timestamps.

13.9 What does NetFlow give you that SNMP does not?

SNMP tells you *how much* traffic crossed an interface. NetFlow tells you *which conversations* made it up — source and destination addresses, ports, protocol, byte and packet counts per flow. For capacity planning, security investigation, and answering "who saturated the link at 02:00," flow data is the only one of the two that can answer the question.

Layer 2 Switching

Access-layer questions are the most practical in the whole set, because this is the layer where a junior engineer will actually be given work — ports, VLANs, port security, and the err-disabled state.

14.1 How does a switch build its MAC address table?

By learning the *source* MAC of every frame it receives and recording it against the ingress port and VLAN. Entries age out after five minutes of inactivity by default. Learning is always from the source address; forwarding decisions are always on the destination address — those two halves are the whole of switching.

14.2 What are the three things a switch can do with a frame?

- **Forward** — the destination MAC is in the table, so send it out that one port.
- **Flood** — the destination is unknown, broadcast, or unknown multicast, so send it out every port in the VLAN except the ingress port.
- **Filter** — the destination is known to be on the same port the frame arrived on, so discard it.

14.3 Store-and-forward, cut-through, or fragment-free?

Store-and-forward receives the entire frame and validates the FCS before forwarding — higher latency, but corrupt frames never propagate. **Cut-through** forwards as soon as it has read the destination MAC, giving minimum latency but relaying errored frames. **Fragment-free** is the compromise: read the first 64 bytes, which catches collision fragments, then forward.

Modern enterprise switches use store-and-forward because it is required for QoS and inspection features, and because ASIC speed has made the latency difference negligible outside high-frequency trading and low-latency data center fabrics.

14.4 What is a CAM table overflow attack?

The attacker floods the switch with frames from thousands of random source MACs until the MAC address table is full. With no room to learn legitimate addresses, the switch floods unicast traffic out every port — effectively turning it into a hub and letting the attacker sniff traffic they should never see.

Port security is the direct countermeasure, since it caps the number of MACs allowed per port.

14.5 Configure port security and explain the violation modes.

```
interface Gi1/0/5
  switchport mode access
  switchport access vlan 20
  switchport port-security
  switchport port-security maximum 2
  switchport port-security violation restrict
  switchport port-security mac-address sticky
```

- **Protect** — silently drop frames from unknown MACs. No log, no counter. Rarely the right choice.
- **Restrict** — drop them, increment a counter, and send a syslog and SNMP trap. Usually the best operational balance.
- **Shutdown** — put the port into err-disabled. The default, and the most secure, but it generates help-desk tickets.

sticky converts dynamically learned addresses into running-configuration entries, so they survive a reload once saved. A maximum of 2 is common on desks where an IP phone and a PC share the port.

14.6 A port is err-disabled. What now?

First find the cause with `show interfaces status err-disabled` and the logs — port security violation, BPDU Guard, a loopback, or a link flap. Fix the cause, then recover the port by shutting and no-shutting it, or configure automatic recovery:

```
errdisable recovery cause psecure-violation
errdisable recovery cause bpduguard
errdisable recovery interval 300
```

WHAT THEY'RE CHECKING

That you diagnose before you recover. Bouncing the port without finding the cause means the same outage returns in five minutes, and that answer is a red flag.

14.7 What is a broadcast storm and how do you limit it?

Broadcast traffic multiplying until it consumes the available bandwidth and switch CPU, usually caused by a Layer 2 loop but sometimes by a faulty NIC or an application misbehaving. Spanning tree prevents the loop case; storm control caps the percentage of broadcast, multicast, or unknown-unicast traffic a port will accept and takes action beyond the threshold.

```
interface Gi1/0/5
storm-control broadcast level 2.00
storm-control action trap
```

14.8 You see repeated MAC flapping messages. What does that mean?

The same MAC address is being learned on two different ports in quick alternation, which almost always means there is a Layer 2 loop. Other possibilities are a mis-paired EtherChannel, a duplicated MAC from a badly cloned virtual machine, or a wireless client roaming between access points. The log tells you exactly which two ports to investigate.

14.9 What is EtherChannel, and how do you negotiate it?

EtherChannel bundles up to eight physical links into one logical interface, so spanning tree sees a single link and does not block any member. That gives you both aggregated bandwidth and sub-second failover.

Negotiation uses **LACP** (802.3ad, open standard, modes active and passive) or **PAgP** (Cisco, modes desirable and auto), or the links can be forced `on` with no protocol at all. Two passive or two auto sides will never form a bundle. All members must agree on speed, duplex, VLAN or trunk mode, and allowed VLANs.

```
interface range Gi1/0/23-24
channel-protocol lacp
channel-group 1 mode active
interface Port-channel1
switchport mode trunk
show etherchannel summary
```

14.10 What is the difference between an SVI and a routed port?

An SVI is a virtual Layer 3 interface for a VLAN, used for inter-VLAN routing and switch management; it is up as long as at least one access port in the VLAN is up. A routed port is a physical port taken out of switching entirely with `no switchport`, given an IP directly, and used for uplinks between Layer 3 devices.

14.11 What does "wire speed" mean, and why can a switch achieve it?

Wire speed means forwarding at the full line rate of every port simultaneously, with no drops caused by the switch itself. Switches achieve it because forwarding is done in purpose-built ASIC hardware rather than by a general-purpose CPU. The CPU handles only the control plane — spanning tree, routing protocols, management — which is exactly why a control-plane attack can bring down a switch that is still forwarding data-plane traffic perfectly.

Spanning Tree Protocol

STP is the most-asked Layer 2 topic and the one candidates most often half-know. Learn the election sequence in order, know the timers, and be ready to explain why a loop is catastrophic rather than merely inefficient.

15.1 What problem does STP solve?

Redundant Layer 2 paths create loops, and Ethernet frames have no TTL to stop them. A single broadcast entering a loop circulates forever, multiplying at every switch — that is a broadcast storm. Two further symptoms follow: MAC table instability as the same address is learned on alternating ports, and multiple copies of unicast frames delivered to hosts.

STP builds a loop-free logical tree over a physically redundant topology by blocking specific ports until they are needed.

WHAT THEY'RE CHECKING

Say "frames have no TTL at Layer 2." That one sentence explains why a Layer 2 loop is fatal while a Layer 3 loop is merely wasteful, and it is the reason STP has to exist at all.

15.2 How is the root bridge elected?

By the lowest bridge ID, which is the bridge priority (default 32768) followed by the MAC address. Since priorities are equal out of the box, the oldest switch — with the lowest MAC — usually wins, which is almost never the switch you want as root.

Always set it deliberately. Priority is configured in increments of 4096 because the extended system ID consumes the lower bits.

```
spanning-tree vlan 1-100 root primary      ! macro: sets priority 24576
spanning-tree vlan 1-100 priority 4096    ! or explicit
```

15.3 What is the port role decision sequence?

Each non-root switch chooses one root port, using these tie-breakers in order:

- Lowest cumulative cost to the root bridge
- Lowest sender bridge ID
- Lowest sender port priority
- Lowest sender port ID

On every segment, the port with the better path becomes designated and forwards; the other becomes non-designated and blocks. All ports on the root bridge are designated.

15.4 What are the default STP port costs?

LINK SPEED	COST	LINK SPEED	COST
10 Mbps	100	1 Gbps	4
100 Mbps	19	10 Gbps	2

Note that cost is accumulated on *ingress* — a switch adds the cost of the port on which the BPDU arrives, not the cost of the port it sends out.

15.5 List the 802.1D port states and timers.

STATE	DURATION	LEARNS MACS?	FORWARDS?
Blocking	Max age, 20 s	No	No — receives BPDUs only
Listening	Forward delay, 15 s	No	No
Learning	Forward delay, 15 s	Yes	No
Forwarding	—	Yes	Yes

That gives roughly 30 to 50 seconds of convergence, which is unacceptable for a modern network — and is precisely why Rapid STP exists. BPDUs are sent every 2 seconds by the hello timer.

15.6 What does Rapid PVST+ change?

802.1w reduces convergence from tens of seconds to typically under a second. It collapses the states into discarding, learning, and forwarding; adds the **alternate** port role (a backup path to the root) and the **backup** role (a redundant path to the same segment); treats point-to-point links specially with a proposal-agreement handshake instead of waiting on timers; and lets every switch originate BPDUs rather than only relaying the root's.

It also builds in the behaviors that were bolted onto 802.1D as UplinkFast and BackboneFast, and its edge-port concept is PortFast by another name.

15.7 PVST+, Rapid PVST+, and MST — what is the difference?

PVST+ runs a separate spanning tree instance per VLAN, which allows per-VLAN load balancing but costs CPU and memory at scale. **Rapid PVST+** is the same model with 802.1w convergence, and is the sensible default on Cisco campus networks. **MST** (802.1s) maps many VLANs onto a small number of instances, so a thousand VLANs can run on two instances instead of a thousand — the right choice in large or multi-vendor environments.

15.8 What is PortFast and where do you use it?

PortFast moves an access port straight to forwarding, skipping listening and learning, so a PC does not wait 30 seconds for DHCP. Use it on every port that connects to an end host — and never on a port connecting to another switch, because you have removed the delay that protects against a loop.

PortFast should always be paired with BPDU Guard, so that if a switch is ever plugged into that port, the port is shut down rather than allowed to form a loop.

15.9 Compare BPDU Guard, Root Guard, and Loop Guard.

FEATURE	TRIGGER	ACTION	WHERE
BPDU Guard	Any BPDU received on a PortFast port	Err-disable the port	Access ports facing users
Root Guard	A superior BPDU received	Root-inconsistent state until it stops	Ports facing switches you do not administer
Loop Guard	BPDUs stop arriving on a non-designated port	Loop-inconsistent state instead of forwarding	Root and alternate ports

The distinction to state clearly: BPDU Guard blocks any switch at the edge, Root Guard allows a switch but not a better root, and Loop Guard protects against unidirectional link failures where BPDUs quietly stop.

15.10 Why should the root bridge be chosen deliberately?

Because the root determines the shape of every path in the topology. If a slow access-layer switch under someone's desk wins the election, traffic between two core switches may traverse it, creating a bottleneck and an unnecessary failure domain. Place the root on a distribution or core switch, and place the secondary root on its peer so that failover is predictable.

15.11 How do you achieve load balancing with STP?

With PVST+ you make one switch root for the odd VLANs and its peer root for the even VLANs, so both uplinks carry traffic instead of one sitting idle in blocking. Per-port cost or priority tuning achieves the same effect more surgically. EtherChannel is the cleaner modern answer where the links go to the same pair of devices, since all members forward.

15.12 Which commands do you use to verify STP?

```
show spanning-tree vlan 20
show spanning-tree root
show spanning-tree interface Gi1/0/1 detail
show spanning-tree inconsistentports
debug spanning-tree events          ! carefully, on a lab device
```

Read `show spanning-tree vlan 20` from the top: is this switch the root, which port is the root port, and which ports are blocking? Those three facts describe the whole topology from this switch's point of view.

VLANs & Trunking

VLANs are the backbone of network segmentation, so this block matters doubly for a security-track role. Expect the native VLAN and VLAN hopping questions — they are the ones that separate candidates.

16.1 What is a VLAN and what does it give you?

A VLAN is a logically separate broadcast domain on shared physical infrastructure. Two ports on the same switch in different VLANs cannot reach each other at Layer 2, no matter that they are inches apart.

The benefits are containment of broadcast traffic, segmentation for security and policy, and flexibility — you can move a user between segments by changing one line of configuration rather than re-cabling. Traffic between VLANs must pass a Layer 3 device, which is precisely where you get to apply an access list.

16.2 Access port or trunk port?

An access port belongs to exactly one VLAN and sends untagged frames to an end device. A trunk carries many VLANs between switches, tagging each frame so the far end knows where it belongs. A PC should never see a tag; a switch-to-switch link almost always should.

```
interface Gi1/0/10
  switchport mode access
  switchport access vlan 20
  switchport nonegotiate
interface Gi1/0/24
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 20,30,40
```

16.3 Describe the 802.1Q tag.

Four bytes inserted into the Ethernet header after the source MAC: a two-byte tag protocol identifier of 0x8100, then three bits of priority (PCP, used for QoS at Layer 2), one drop-eligible bit, and twelve bits of VLAN ID. Twelve bits give 4096 values, of which 1 through 4094 are usable.

Because the frame grows by four bytes, a trunk carrying full-size frames needs to accept 1504 bytes at Layer 2 — the origin of the "baby giant" frame.

16.4 What is the native VLAN?

The one VLAN on a trunk whose frames are sent untagged, VLAN 1 by default. It exists for backward compatibility with devices that do not understand tagging, and it carries control traffic such as CDP and DTP.

Best practice is to change it to an unused VLAN, make sure it matches on both ends, and never use it for user data. A mismatch does not bring the link down — it silently merges two VLANs, which is both a hard-to-find outage and a security hole.

16.5 Explain VLAN hopping.

Two variants:

- **Switch spoofing** — the attacker's host pretends to be a switch and negotiates a trunk using DTP, gaining access to every VLAN. Prevented by hard-coding access ports and disabling DTP with `switchport nonegotiate`.
- **Double tagging** — the attacker sends a frame with two tags. The first switch strips the outer tag, which matches the native VLAN, and forwards the frame with the inner tag still in place, delivering it into the target VLAN. Prevented by never using the native VLAN for user traffic and by pruning unused VLANs from trunks.

WHAT THEY'RE CHECKING

Double tagging only works when the attacker's access VLAN is the trunk's native VLAN. Saying that condition out loud proves you understand the mechanism rather than the label.

16.6 Why is leaving everything in VLAN 1 discouraged?

VLAN 1 cannot be deleted, is the default for every port, is the default native VLAN on every trunk, and carries control-plane protocols. Using it for user data means an attacker who reaches any unconfigured port lands in the same broadcast domain as your switch management traffic. The convention is to move users into purpose-built VLANs, put management in its own VLAN, and set the native VLAN to an unused ID.

16.7 What is a voice VLAN?

A configuration that lets one access port carry two VLANs: untagged data for the PC plugged into the phone, and tagged voice for the phone itself. The switch tells the phone which VLAN and CoS to use via CDP or LLDP-MED. It keeps voice separate for QoS and security without requiring a second cable to every desk.

```
interface Gi1/0/10
  switchport mode access
  switchport access vlan 20
  switchport voice vlan 110
  spanning-tree portfast
```

16.8 How do you route between VLANs?

Three options, in order of how you will meet them in the field:

- **SVIs on a multilayer switch** — one virtual interface per VLAN, routing in hardware. The standard campus design.
- **Router on a stick** — one physical link carrying a trunk, with a subinterface and `encapsulation dot1q` per VLAN. Cheap, but all inter-VLAN traffic shares that one link.
- **Separate physical interfaces** — one router port per VLAN. Simple, and it does not scale past a handful.

```
interface Gi0/0.20
  encapsulation dot1q 20
  ip address 10.20.0.1 255.255.255.0
interface Gi0/0.999
  encapsulation dot1q 999 native
```

16.9 Why restrict the allowed VLAN list on a trunk?

Because by default a trunk carries all 4094 VLANs, which means broadcast traffic from VLANs that have no members on the far switch is being flooded across the link for no reason. Pruning the list reduces unnecessary traffic, shrinks the blast radius of a misconfiguration, and removes the raw material for double-tagging attacks.

16.10 Two hosts in the same VLAN on different switches cannot communicate. Where do you look?

In order: does the VLAN exist in the VLAN database on *both* switches; are both access ports actually in that VLAN; is the link between them a trunk on both ends; is the VLAN in the allowed list on that trunk; and is any port blocking due to spanning tree? `show vlan brief`, `show interfaces trunk`, and `show spanning-tree vlan X` answer all five.

A VLAN that exists on one switch but not the other is the single most common cause, and it produces exactly this symptom.

16.11 What is the difference between the normal and extended VLAN ranges?

Normal range is 1 to 1005, stored in the `vlan.dat` file and propagated by VTP versions 1 and 2. Extended range is 1006 to 4094, is not advertised by older VTP versions, and requires the switch to be in transparent mode — or running VTPv3 — to be configured. VLANs 1002 through 1005 are reserved for legacy Token Ring and FDDI.

16.12 What is DTP and should you use it?

The Dynamic Trunking Protocol negotiates whether a link becomes a trunk. It is convenient and it is an attack surface, since a host running the right software can negotiate a trunk for itself. Disable it everywhere with `switchport nonegotiate` and configure trunks and access ports explicitly. Deterministic configuration beats automatic negotiation on any link that matters.

VTP

VTP is the rare topic where the best interview answer is "here is how it works, and here is why I would not run it." Say both halves.

17.1 What is VTP?

The VLAN Trunking Protocol synchronizes the VLAN database across switches in a common VTP domain, so a VLAN created once on a server is learned everywhere. It advertises only VLAN existence and names — it never distributes port assignments, and it does not create trunks despite the name.

17.2 Name the VTP modes.

- **Server** — can create, modify, and delete VLANs, and advertises them. The default.
- **Client** — cannot change VLANs locally; accepts and forwards advertisements.
- **Transparent** — maintains its own local database, ignores advertisements, but still forwards them to other switches.
- **Off** — does not participate or forward at all.

17.3 How does the revision number work?

Every change to the VLAN database on a server increments a revision number. Switches accept any advertisement with a higher revision than their own and overwrite their database with it. There is no notion of "better" or "newer by timestamp" — only higher.

17.4 What is the classic VTP disaster?

A switch with a high revision number and a nearly empty VLAN database is connected to the production network — typically a lab switch, or a returned unit that someone reset incorrectly. Because its revision is higher, every switch in the domain accepts its database and deletes the real VLANs. Every access port lands in a nonexistent VLAN and the network goes dark in seconds.

The safe practices: put spare and lab switches in transparent or off mode, set a VTP password on the domain, and reset the revision counter by moving a switch to transparent and back before connecting it.

WHAT THEY'RE CHECKING

This is a war story question. Answering it with the failure mode and the prevention shows judgment, not just recall — and most interviewers have personally lived through it.

17.5 What must match for VTP to work?

The domain name is case sensitive and must match; the password, if set, must match; the VTP version must be compatible; and the link between switches must be a trunk, since advertisements are only sent over trunks.

17.6 Compare VTP versions 1, 2, and 3.

Version 1 and 2 are similar, with v2 adding Token Ring support and consistency checking. Version 3 is the significant one: it supports extended-range VLANs, introduces a **primary server** role so that only one explicitly promoted switch can make changes, protects against accidental overwrites, and can also propagate MST and private VLAN configuration. If VTP is required, v3 is the only version worth deploying.

17.7 What is VTP pruning?

It stops flooding broadcast and unknown-unicast traffic for a VLAN down trunks toward switches that have no active ports in that VLAN. It saves bandwidth. Manually setting the allowed VLAN list on each trunk achieves the same result with more predictability and no dependence on VTP.

17.8 Would you deploy VTP today?

Usually not. The administrative saving is small — VLANs are created rarely — while the failure mode takes down an entire campus. Most organizations run transparent or off mode with VLANs defined by automation or configuration templates, which are auditable and version-controlled. If a large environment genuinely needs central VLAN management, use VTPv3 with a designated primary server and a password.

WAN Technologies

Much of the classic WAN syllabus is now historical. Know the legacy vocabulary well enough to discuss it, but be ready to pivot to what actually connects branches today: Metro Ethernet, MPLS, broadband, and SD-WAN.

18.1 Distinguish LAN, MAN, and WAN.

A LAN covers a building or campus and is normally owned by the organization. A MAN spans a city, typically as Metro Ethernet from a provider. A WAN spans regions or countries and uses infrastructure you lease rather than own. The practical difference is not distance but ownership: on a WAN, bandwidth costs money per month and latency is set by geography, so design decisions change accordingly.

18.2 Explain the WAN terms CPE, CO, DCE, and DTE.

The CPE is the customer premises equipment at your site; the CO is the provider's central office; the local loop connects the two. The **DCE** provides clocking and is normally the provider's CSU/DSU, while the **DTE** is your router receiving that clock. In a back-to-back lab serial connection you must supply the clock yourself with `clock rate` on the DCE end — a classic lab gotcha.

18.3 HDLC or PPP?

Cisco HDLC is the default serial encapsulation on IOS and is proprietary in that Cisco added a type field to support multiple protocols. It has no authentication and no negotiation.

PPP is the open standard and adds real functionality: LCP for link establishment and options, NCP for multiple network-layer protocols, authentication with PAP or CHAP, multilink bonding, error detection, and compression. Any link between different vendors, or any link needing authentication, uses PPP.

18.4 PAP or CHAP?

PAP sends the username and password in cleartext, once, at link setup. CHAP uses a three-way challenge-response with an MD5 hash of a random challenge, never transmits the password, and re-challenges periodically during the session. Choose CHAP; PAP exists only for interoperability with equipment that cannot do better.

18.5 What was Frame Relay, and what is a DLCI?

Frame Relay was a packet-switched WAN service that let one physical circuit carry many virtual circuits to different sites, sharing capacity statistically instead of paying for dedicated bandwidth per destination. The **DLCI** is a 10-bit, locally significant identifier for one virtual circuit — the two ends of the same PVC usually have different DLCI numbers.

It is a non-broadcast multi-access network, which is why it needed inverse ARP for mapping and why split horizon caused so much trouble in hub-and-spoke designs. It has been almost entirely replaced by MPLS and Ethernet services.

18.6 What is MPLS and why did providers adopt it?

MPLS forwards packets on a short label rather than a full routing lookup, with labels pushed at the ingress provider edge, swapped through the core, and popped at the egress. That gave providers fast forwarding, traffic engineering, and — most importantly — the ability to run many customers' L3VPNs over one shared core while keeping their routing tables separate.

From the customer's side, an MPLS L3VPN behaves like a private cloud: you peer your CE router with the provider's PE and every site appears directly connected.

18.7 Compare a site-to-site VPN with a remote-access VPN.

A site-to-site tunnel joins two networks permanently, is configured once on the two gateways, and is invisible to users.

Remote access connects one user's device to the network on demand, using a client or a browser-based SSL VPN, and requires per-user authentication and posture checking.

Site-to-site is a network design decision; remote access is an identity and endpoint decision.

18.8 Which protocols build a site-to-site tunnel?

IPsec provides confidentiality, integrity, authentication, and anti-replay, using IKE for key exchange and ESP for encryption. It cannot natively carry multicast or routing protocols in tunnel mode.

GRE encapsulates almost anything, including multicast and routing protocol traffic, but offers no security — which is why **GRE over IPsec** is such a common combination. **DMVPN** extends the idea with dynamic spoke-to-spoke tunnels built on NHRP, so a hub-and-spoke design does not have to hairpin all traffic through the hub.

18.9 What is SD-WAN?

An overlay architecture that separates the control plane from the forwarding plane and manages branch connectivity centrally. Each site can use several transports at once — MPLS, broadband, LTE — and the controller applies policy that steers each application over the best available path based on measured loss, latency, and jitter.

The reason it took over is economic as much as technical: it lets a business replace expensive MPLS circuits with commodity internet for most traffic, with encryption and path selection compensating for the loss of a private circuit.

18.10 Why does tunneling create MTU problems?

Because every layer of encapsulation adds bytes. GRE adds 24, IPsec ESP adds more depending on the cipher and mode, and the result no longer fits inside the underlying 1500-byte MTU. The packet must then be fragmented — or dropped, if DF is set and ICMP is filtered, which produces the black hole where pings and web pages work but large transfers hang.

The standard remedies are to lower the tunnel MTU, clamp the TCP MSS with `ip tcp adjust-mss`, and make sure ICMP type 3 code 4 is permitted end to end.

Wireless

Wireless questions test whether you understand that RF is a shared, half-duplex, interference-prone medium. Almost every wireless troubleshooting answer comes back to that one fact.

19.1 Summarize the 802.11 standards.

STANDARD	ALSO CALLED	BAND	NOTES
802.11b / g	—	2.4 GHz	11 and 54 Mbps, legacy
802.11a	—	5 GHz	54 Mbps, contemporary with b
802.11n	Wi-Fi 4	2.4 and 5 GHz	MIMO and channel bonding
802.11ac	Wi-Fi 5	5 GHz	Wider channels, MU-MIMO downlink
802.11ax	Wi-Fi 6 / 6E	2.4, 5, and 6 GHz	OFDMA, better efficiency in dense areas

The theme across generations is efficiency in crowded environments, not just raw speed. Wi-Fi 6's real contribution is serving many clients at once rather than making one client faster.

19.2 2.4 GHz or 5 GHz?

2.4 GHz travels further and penetrates walls better, but has only three non-overlapping 20 MHz channels — 1, 6, and 11 — and shares the band with Bluetooth, microwaves, and older cordless devices. 5 GHz has far more channels and much less interference, at the cost of range.

Design consequence: use 5 GHz as the primary band for clients and keep 2.4 GHz for legacy and IoT devices. And never use channels other than 1, 6, and 11 on 2.4 GHz, because a partially overlapping channel is worse than a shared one.

19.3 Define SSID, BSSID, BSS, and ESS.

The **SSID** is the network name. The **BSSID** is the MAC address of a specific radio on a specific AP, so it identifies one cell. A **BSS** is that single cell; an **ESS** is multiple APs sharing one SSID over a distribution system so clients can roam between them without changing networks.

19.4 Autonomous or lightweight APs?

An autonomous AP is self-contained and configured individually — acceptable for a handful of APs, unmanageable for hundreds. A lightweight AP registers with a wireless LAN controller over a CAPWAP tunnel and receives its configuration centrally.

This is the **split-MAC** architecture: the AP keeps the real-time functions — transmitting, acknowledging frames, encryption, beaconing — while the controller handles association, authentication, RF management, roaming coordination, and policy. Centralizing those makes coordinated channel and power planning possible in the first place.

19.5 Compare WEP, WPA, WPA2, and WPA3.

WEP used RC4 with a fatally short IV and is broken in minutes. WPA was the transitional fix using TKIP on the same hardware. WPA2 mandated AES-CCMP and remained the standard for over a decade, though it is vulnerable to offline dictionary attack against a captured handshake if the passphrase is weak. WPA3 replaces the pre-shared key handshake with SAE, which resists offline cracking, adds forward secrecy, and provides opportunistic encryption for open networks.

Never deploy WEP or TKIP. WPA2-AES is the minimum acceptable, WPA3 the target.

19.6 Personal or Enterprise mode?

Personal uses one pre-shared key for everyone, so changing it means retouching every device, and any departing employee still knows it. Enterprise uses 802.1X with a RADIUS server and per-user credentials or certificates via EAP, which gives individual authentication, per-user revocation, dynamic VLAN assignment, and an audit trail.

Any corporate network with more than a handful of users should be running Enterprise; PSK belongs on the guest and IoT networks.

WHAT THEY'RE CHECKING

For security-track roles, mention that 802.1X involves three parties — supplicant, authenticator, and authentication server — and that the same model secures wired ports.

19.7 What is a rogue AP, and what is an evil twin?

A rogue AP is any unauthorized access point on your network — often an employee's own device plugged in for convenience, which bypasses your entire wireless security policy. An evil twin is deliberately hostile: an AP broadcasting your SSID, often with a stronger signal, to lure clients into associating so the attacker can harvest credentials or intercept traffic.

Detection comes from controller-based rogue scanning and wireless intrusion prevention; prevention on the wired side comes from 802.1X port authentication and DHCP snooping.

19.8 Why is wireless half-duplex, and why does that matter?

Because all stations share one frequency and a radio cannot transmit and receive on it simultaneously. Every station, including the AP, must contend for airtime, so the advertised data rate is shared among all clients on the channel — and the slowest client consumes a disproportionate share of airtime while it transmits.

This is why adding more APs on the same channel does not add capacity, and why co-channel interference is the dominant cause of poor wireless performance in dense deployments.

19.9 Users report slow Wi-Fi in one area. How do you approach it?

Separate coverage from capacity from interference. Check the client's signal strength and signal-to-noise ratio, not just the bar count; check how many clients and how much retry traffic are on that channel; look for co-channel interference from your own APs and for non-Wi-Fi sources on 2.4 GHz; check whether clients are stuck on 2.4 GHz or clinging to a distant AP instead of roaming.

The correct fix is often to *reduce* transmit power so cells are smaller and roaming happens sooner — which sounds counterintuitive and is exactly why the question is asked.

19.10 What is a site survey?

A structured RF assessment of a physical space: a predictive model from floor plans first, then an on-site passive and active survey measuring actual signal, noise, interference, and throughput. It produces AP placement, channel plan, power levels, and antenna choices.

Skipping it is how organizations end up with too many APs on overlapping channels, which performs worse than fewer APs placed properly.

First Hop Redundancy Protocols

A short block with a clean logic: a host can only have one default gateway, so redundancy has to be created below the host's awareness. Know which protocol is proprietary and which is not.

20.1 What problem does an FHRP solve?

A host has exactly one default gateway address. If that router fails, every host on the subnet loses off-subnet connectivity — regardless of how redundant the rest of your network is. An FHRP lets two or more routers share a single virtual IP and virtual MAC address, so the hosts keep using the same gateway while the physical device behind it changes.

Sharing the virtual MAC as well as the IP is the essential detail: it means hosts do not need to re-ARP, so failover is invisible to them.

20.2 Compare HSRP, VRRP, and GLBP.

	HSRP	VRRP	GLBP
Origin	Cisco	Open standard (RFC 5798)	Cisco
Roles	Active / Standby	Master / Backup	AVG / AVF
Load sharing	Only via multiple groups	Only via multiple groups	Native, per host
Default priority	100	100	100
Timers	Hello 3 s, hold 10 s	Advert 1 s, master down 3 s	Hello 3 s, hold 10 s

Choose VRRP in a multi-vendor environment, HSRP in an all-Cisco one, and GLBP when you want both gateways carrying traffic without maintaining several groups.

20.3 Give me the HSRP and VRRP addressing details.

HSRP version 1 uses multicast 224.0.0.2 on UDP 1985 and supports groups 0 to 255; version 2 uses 224.0.0.102, supports groups up to 4095, and adds IPv6 and millisecond timers. The HSRP virtual MAC is 0000.0c07.ac followed by the group number in hex.

VRRP uses IP protocol 112 to multicast 224.0.0.18, and its virtual MAC is 0000.5e00.01 followed by the group ID. VRRP also allows the virtual IP to be the real address of one router, which then takes priority 255 as the address owner.

20.4 How is the active router chosen, and what is preemption?

Highest priority wins, with the highest interface IP address as the tie-breaker. Preemption decides what happens when a higher-priority router appears *after* the election: with `standby 1 preempt` it takes over, and without it the current active router keeps the role until it fails.

HSRP does not preempt by default and VRRP does — a difference worth stating, because it explains why a network can behave differently after a reload than the diagram suggests.

20.5 Configure HSRP.

```
interface Vlan20
 ip address 10.20.0.2 255.255.255.0
 standby version 2
 standby 20 ip 10.20.0.1           ! the virtual IP hosts point at
 standby 20 priority 110
 standby 20 preempt delay minimum 60
 standby 20 authentication md5 key-string HsrpSecret
 standby 20 track 1 decrement 30
 ! verification
 show standby brief
```

The `preempt delay` matters in production: without it, a router that has just rebooted can seize the active role before its routing protocol has converged, black-holing traffic for several seconds.

20.6 Why is interface or object tracking necessary?

Because an FHRP only monitors the LAN-facing interface. If the active router's *uplink* fails while its LAN port stays up, it remains the active gateway and cheerfully black-holes every packet. Tracking an uplink interface or an IP SLA reachability object, and decrementing priority when it fails, is what makes failover actually correlate with reachability.

WHAT THEY'RE CHECKING

This is the best question in the FHRP block, because it distinguishes candidates who configured HSRP in a lab from those who have thought about how it fails.

20.7 How do you load-share with HSRP or VRRP?

Create two groups on the same subnet with two virtual IPs, make router A active for group 1 and router B active for group 2, then point half the hosts at each virtual IP via DHCP. It works and it is common. GLBP does the same thing automatically by handing out different virtual MACs in response to ARP requests.

20.8 What happens on the wire when failover occurs?

The standby stops hearing hellos, waits out the hold time, promotes itself, and sends a gratuitous ARP announcing the virtual MAC from its own port. Hosts do not need to change anything because the IP and MAC are unchanged — the switches simply relearn which port that MAC lives on. Existing TCP sessions typically survive, since the outage is on the order of a few seconds.

Answers People Get Wrong

These errors circulate widely in copied question banks and study notes. Each one is a fast way for an interviewer to identify a candidate who memorized a document instead of learning the material.

~~DoS stands for Disk Operating System.~~

In networking, **DoS is denial of service**. The disk operating system meaning belongs to 1980s PC history and has no place in an infrastructure interview.

~~Flooding is when a router forwards a packet to every attached node.~~

Flooding is a **switch** behavior at Layer 2, applied to broadcasts, unknown unicasts, and unknown multicasts within a VLAN. Link-state protocols also "flood" LSAs, but that is a distinct mechanism with its own rules — do not merge the two definitions.

~~The OSI model was released in February 1980, which is why the standards are numbered 802.~~

Two different bodies. The **IEEE 802** project takes its number from its February 1980 start date and produces Ethernet and Wi-Fi standards. The **OSI reference model** is ISO/IEC 7498, published in 1984. The naming coincidence is not a causal relationship.

~~The frame check sequence is a function of the LLC sublayer.~~

The FCS belongs to the **MAC sublayer**, along with framing, addressing, and media access. LLC handles identification of the upper-layer protocol and optional flow control.

~~The session layer keeps data from different applications separate.~~

That is the **transport layer**, using port numbers to multiplex conversations. The session layer establishes, maintains, and tears down dialogs — a related but different job.

~~A class C address supports 254 hosts.~~

A **/24 prefix** supports 254 usable hosts. Class C describes the range 192.0.0.0 through 223.255.255.255 — roughly two million separate networks. Classful language has been obsolete since CIDR, so answer in prefix lengths and mention classes only as history.

~~Cut through is a technique where the router forwards the frame immediately.~~

Cut-through, fragment-free, and store-and-forward are **switching** methods operating on frames at Layer 2. Routers process packets at Layer 3 and must read further into the header than a MAC address.

~~ICMP type 8 is echo packet and type 0 is echo request.~~

Type 8 is echo request; type 0 is echo reply. Getting this backwards in an interview is memorable for the wrong reason, since it is the pair everyone uses daily.

~~Switches cannot be used in large networks because they cannot forward between subnets.~~

A basic Layer 2 switch cannot, but **multilayer switches route** — and they are the backbone of essentially every modern campus, handling inter-VLAN routing in hardware faster than most routers.

~~AD means either administrative distance or advertised distance, interchangeably.~~

Context decides. In route selection, **AD is administrative distance**, the trust level of a route source. In EIGRP, the neighbor's own metric is properly called the **reported distance**, though older documentation says advertised distance. Say which one you mean rather than relying on the abbreviation.

Subnetting Under Pressure

You will be asked to subnet on a whiteboard, without a calculator, while someone watches. This is the minimum you need in memory and the method that survives nerves.

The one table to memorize

PREFIX	MASK	BLOCK SIZE	USABLE HOSTS	SUBNETS PER /24
/24	255.255.255.0	256	254	1
/25	255.255.255.128	128	126	2
/26	255.255.255.192	64	62	4
/27	255.255.255.224	32	30	8
/28	255.255.255.240	16	14	16
/29	255.255.255.248	8	6	32
/30	255.255.255.252	4	2	64

The only mask octet values that ever appear are 128, 192, 224, 240, 248, 252, 254, and 255. If you write anything else, you have made an arithmetic error.

The three-step method

1. Find the block size. Subtract the interesting octet of the mask from 256. For /26, that is $256 - 192 = 64$.

2. Count in blocks until you pass the address. For 172.16.20.100/26: 0, 64, 128. The address 100 falls in the block starting at 64.

3. Read off the answer. Network 172.16.20.64, broadcast 172.16.20.127, usable range .65 to .126.

Say the steps out loud as you work. Interviewers are grading your method as much as your answer — and a visible method means a wrong arithmetic step still earns credit.

Drills — cover the answers

QUESTION	ANSWER
Network and broadcast for 10.5.5.130/25?	10.5.5.128 and 10.5.5.255
How many /28 subnets fit in a /24?	16
Smallest prefix for 500 hosts?	/23, giving 510 usable
Usable range in 192.168.4.48/29?	192.168.4.49 to .54

A VLSM example worth rehearsing

Given 192.168.1.0/24, allocate subnets for 60, 30, 12, and 2 hosts with no waste. Always assign the largest requirement first.

REQUIREMENT	PREFIX	SUBNET	USABLE RANGE
60 hosts	/26	192.168.1.0	.1 – .62
30 hosts	/27	192.168.1.64	.65 – .94
12 hosts	/28	192.168.1.96	.97 – .110
2 hosts (WAN link)	/30	192.168.1.112	.113 – .114

Everything from 192.168.1.116 onward remains free for growth — which is the point of doing it largest-first.

Before the interview

Pick five questions from this handbook at random and answer them out loud, timed. If any answer runs past ninety seconds, it is too long — cut it to the definition, the mechanism, and one example. Then build the topology for one section in a lab and break it deliberately, because the questions that decide interviews are the ones about what happened when something failed.

And prepare two of your own: what the team is currently struggling with, and what the first ninety days look like. Candidates who ask those are remembered.

Prepared for OAK Academy students. The topic scope follows a widely circulated CCNA interview question outline; all explanations, examples, and configurations here are original and were written and technically reviewed for this edition. Commands target Cisco IOS and IOS-XE and should be verified against the platform and version you are running.